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个人简介

目前就职于中国科学院计算技术研究所，是处理器芯片全国重点实验室的副研究员，主要研究方向为专用芯片与系统设计、智能处理器设计、容错计算等，主持来自中科院先导专项、科技部重点研发、军工专项、基金委以及航天科技集团等多个相关的研究项目，将专用芯片以及硬件加速器应用于星载智能计算、舰载大数据处理、无人机等多个关键场景。发表集成电路设计相关的学术论文近 100 篇，其中通信/一作 50 余篇，出版英文专著 1 部、书籍章节 2 个，获得 **2021 年 IEEE Transaction on Computers（体系结构顶刊，CCF-A）年度唯一最佳论文奖**以及 GLVLSI 国际会议最佳论文奖，研究工作也获得了产业界认可，获得 2019 年中科院科技成果转化特等奖以及 **2023 年华为奥林帕斯先锋奖（国际范围内评选，每年 4 项）**。指导学生获得多个国际竞赛前三的奖项，其中包括 ICCAD 的 EDA 竞赛一等奖、DAC 系统设计竞赛二、三等奖、CVPR 低功耗计算机视觉挑战赛三等奖、PPoPP 2025 图计算竞赛三等奖。

担任 IEEE Transactions on Emerging Topics in Computing (TETC) 编委，IEEE Transactions on Parallel and Distributed Systems (TPDS) 评审委员，长期担任 DAC、FPT、ICLR、NeurIPS、ICLR、DFTS 等国际学术会议的程序委员，以及领域内多个期刊如 TCAD、TVLSI、TPDS 等的审稿人，目前是 IEEE 高级会员以及 CCF 高级会员。多次获得中科院计算所优秀员工称号，2023 年参加中组部青年拔尖人才计划竞争，是所内唯二（共 18 人参选）进入当年会评的申请人。

教育与工作经历

中科院计算所 *2018-06 - 至今*

计算机体系结构国家重点实验室/处理器芯片全国重点实验室，副研究员

新加坡国立大学 *2016-12 - 2018-06*

计算机学院，博后

香港大学 *2011-09 - 2016-11*

计算机工程专业，电机与电子工程系

哈尔滨工业大学 *2007-09 - 2009-07*

微电子学与固体电子学专业，航天学院

哈尔滨工业大学 *2003-09 - 2007-07*

电子信息科学与技术专业，航天学院

科研项目

跨层芯片设计与优化软硬件平台 *2024 - 2028*

中科院先导 B 计划课题，1100 万，主持

基于智能计算存储器的大数据处理系统设计 *2023 - 2024*

专项项目，130 万，主持

领域专用芯片跨层参数优化方法 *2023 - 2025*

国家重点研发计划子课题，160 万，主持

弹性容错深度学习处理器设计 *2022 - 2025*

自然科学基金面上项目，57 万，主持

基于 FPGA 的高能效图计算加速器设计 *2020 - 2022*

自然科学基金青年项目，27 万，主持

面向遥感图像识别的智能微系统可靠性评测 *2024 - 2024*

航天科技集团，50 万，主持

面向 COTS 器件的容错深度学习工具链设计 *2023 - 2023*

航天科技集团，30 万，主持

AI 辅助的领域专用芯片设计自动化方法研究 *2023 - 2024*

处理器芯片重点实验室培育项目，30 万，主持

深度学习处理器自动化容错设计方法 *2021 - 2022*

计算机体系结构重点实验室培育项目，30 万，主持

人工智能芯片测评技术研究 *2023 - 2024*

专项项目，290 万，参与

基于分离式内存的近存智能计算架构研究 *2024 - 2025*

专项项目，130 万，参与

基于微存储的计算型存储 *2023 - 2024*

华为，120 万，参与

物端智能无人机系统设计 *2019 - 2020*

项目子课题，600 万，参与

开源物端智能处理器 *2020 - 2021*

中科院先导 C 课题，1200 万，参与

基于智能网卡与智能存储设备的流式计算系统研究 *2020 - 2021*

华为，120 万，参与

论文

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